

NGUYỄN VĂN LÊ BÁ THÀNH

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SUMMARY

A proactive and detail-oriented Computer Vision undergraduate (CPA: 3.6/4.0) with a strong foundation in machine learning theory and hands-on experience in developing end-to-end AI solutions. Passionate about applying AI/ML to solve complex problems, particularly within the healthcare technology sector. Proficient in Python, PyTorch, and TensorFlow, with proven project experience in multimodal deep learning, computer vision, and model deployment on edge devices.

EDUCATION

University of Science – VNU-HCM

Bachelor's Degree in Computer Science, Major in Computer Vision - CPA: 3.6/4.0

Ho Chi Minh City, Viet Nam

Sep 2022 – Aug 2026 (*Expected*)

TECHNICAL SKILLS

Programming Languages: Python, C++/C, Dart.

AI/ML Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib.

Tools & Platforms: Git, Docker, Jupyter Notebook, Visual Studio Code, Overleaf.

LANGUAGES

English: Advanced (IELTS 7.0)

PROJECTS & EXPERIENCE

Multimodal Classification System for Bone Tumor (Healthcare AI)

Aug 2025 – Present

Developing a multimodal deep learning system to enhance bone tumor classification by fusing medical images (X-rays) with clinical text data, directly addressing challenges in healthcare technology.

- Researched state-of-the-art **Generative AI** models, including Vision-Language Models (VLMs), to design a novel fusion architecture for enhanced diagnostic accuracy.
- Engineered a robust data pipeline, implementing techniques for **data preprocessing** and augmentation for both image (DICOM) and text (clinical notes) modalities.
- Established a comprehensive **model evaluation** framework using metrics such as Accuracy, Precision, Recall, and F1-score to validate system performance against established benchmarks.

Content-Based Image Retrieval System | [GitHub Link](#)

May 2025 – Aug 2025

Built a high-performance image retrieval system from scratch using classical computer vision algorithms and an optimized indexing strategy in C++.

- Implemented multiple handcrafted feature extractors (SIFT, ORB, Color Histograms) for comprehensive image representation.
- Optimized the forward indexing system to achieve **sub-second (<1000ms) average search times**, demonstrating strong problem-solving skills in performance tuning.

AIoT Smart Greenhouse | [GitHub Link](#)

May 2025 – Aug 2025

Led a team to develop an end-to-end IoT system with an embedded AI model for voice-activated control, showcasing skills in both hardware integration and on-device ML deployment.

- Trained and fine-tuned a lightweight keyword detection model, achieving **98% accuracy** on test data.
- Successfully deployed the optimized model onto a microcontroller (ESP32), enabling real-time, low-latency voice command recognition.
- Practiced effective **time management** and collaboration, leading the team to deliver the full project on schedule.

CERTIFICATES & PROFESSIONAL DEVELOPMENT

Machine Learning in Production (DeepLearning.AI) Certificate	Nov 2025
• Gained proficiency in MLOps concepts, including deploying models, monitoring performance, and addressing production challenges like data drift and concept drift.	
Prepare Data for Exploration (Google) Certificate	Jun 2024
• Mastered techniques for data collection, cleaning, and structuring to ensure data quality and reliability for machine learning applications.	